

Fixed Points III: Disturbance–Damping Balance and Stability

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Part III of VI in the Fixed Point series. As both the cornerstone of the Modal Triplet Theory (MTT) collection and a stand-alone development, the series is intended to function simultaneously as a basis and as a self-contained study. Each paper in the series builds upon its predecessors, extending the fixed-point framework step by step.

Abstract

We rebuild disturbance–damping stability on the joint spectral decomposition of the corrected FP–II internal operators. For each noncoherent joint mode α , the one-sided nonlinear damping margin is $\gamma_\alpha = d_\alpha - L_\alpha$. Deterministic force amplitude f_α and stochastic noise power q_α are distinct quantities. We prove the deterministic input-to-state floor $\limsup |a_\alpha| \leq f_\alpha/\gamma_\alpha$ and the stochastic second-moment floor $\limsup \mathbb{E}|a_\alpha|^2 \leq q_\alpha/(2\gamma_\alpha)$ when $\gamma_\alpha > 0$. Gaussian invariant laws and necessity of the sign condition are asserted only for exact Ornstein–Uhlenbeck dynamics or robust worst-case stability. Bundlewise results apply to $Q\Psi$ and use separate stochastic trace and deterministic weighted-series conditions. Finally, deterministic homogenization is made conditional on an enhanced functional central-limit theorem, tightness, and rough-path convergence; the corrected Green–Kubo tensor is $D = \int_0^\infty (R + R^*) ds$.

Revision note for this edition

Supersedes. *Fixed Points III: Disturbance–Damping Balance and Stability*, version 3.

Reason. Deterministic amplitudes and stochastic powers were conflated, nonlinear systems were assigned Gaussian/OU conclusions, and the Green–Kubo normalization and homogenization hypotheses were incomplete.

Resolution. Version 4 derives separate deterministic and stochastic floors on the joint modal spectrum, restricts Gaussian claims to exact OU dynamics, and states the enhanced invariance-principle and rough-path gates.

Retained result. Positive net modal damping still gives quantitative noncoherent stability and disturbance floors.

Remaining boundary. Coherent disturbances, nonlinear invariant laws, and physical stochastic interpretation require additional assumptions.

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1 Scope and inherited framework

We use the corrected FP–I/II control framework. The compact internal space carries strongly commuting nonnegative self-adjoint operators A_1, A_2, A_3 on one common Hilbert space. Their joint harmonic projector is P , with $Q = I - P$. All stability results below concern $Q\Psi$ unless coherent forcing is explicitly introduced. The stabilization parameter is not physical Lorentzian time.

Deterministic fixed points of a time-step map and stochastic invariant measures are different objects. Existence of the former is inherited from FP–I/II under their invariant-set and compactness/condensing hypotheses. Existence of the latter requires a Markov/Feller and tightness argument stated separately below. For analytic-semigroup and dissipative-flow background, see [1, 6, 10]; for stochastic evolution and exact Ornstein–Uhlenbeck processes, see [9, 4]. The companion analytic framework is [8].

2 Joint modal decomposition

Strong commutation supplies a joint spectral calculus. In the compact pure- point case, choose a joint orthonormal basis $\{e_\alpha\}$ such that

$$A_n e_\alpha = \lambda_\alpha^{(n)} e_\alpha, \quad n = 1, 2, 3.$$

The multi-index α includes all three spectral labels and any multiplicity label. Define

$$d_\alpha := \sum_{n=1}^3 \kappa_n \lambda_\alpha^{(n)}, \quad \Lambda_\alpha := \sum_{n=1}^3 \lambda_\alpha^{(n)}.$$

Let $\mathcal{I}_Q$ contain the indices for which at least one $\lambda_\alpha^{(n)}$ is positive. Then

$$Q\Psi(t) = \sum_{\alpha \in \mathcal{I}_Q} a_\alpha(t) e_\alpha.$$

This joint index prevents multiple counting when the vertical structures overlap. It also applies to a genuinely nested realization after its inclusion maps have been supplied; rank labels alone do not establish nesting.

Assumption 2.1 (One-sided modal remainder bound). On the invariant set, the nonlinear modal remainder satisfies

$$\operatorname{Re}(\overline{a_\alpha} R_\alpha(a)) \leq L_\alpha |a_\alpha|^2, \quad \gamma_\alpha := d_\alpha - L_\alpha.$$

The remainder may couple modes; the displayed one-sided estimate, rather than an entrywise linearization identity, is the hypothesis used in the energy inequalities.

3 Two distinct disturbance classes

3.1 Deterministic forcing

For deterministic input $b_\alpha(t)$ assume

$$\|b_\alpha\|_{L^\infty(0,\infty)} \leq f_\alpha,$$

where f_α has units of amplitude per stabilization time. The mode equation is

$$\dot{a}_\alpha = -d_\alpha a_\alpha + R_\alpha(a) + b_\alpha(t). \quad (3.1)$$

3.2 Stochastic forcing

For the scalar real stochastic theorem let W_α be standard Brownian motions and let $q_\alpha \geq 0$ be noise powers. The mode equation is

$$da_\alpha = (-d_\alpha a_\alpha + R_\alpha(a)) dt + \sqrt{q_\alpha} dW_\alpha(t). \quad (3.2)$$

Correlated or infinite-dimensional noise is described by a positive covariance operator Q_ξ ; then the scalar q_α are replaced by its matrix entries and the bundlewise criterion is a weighted trace condition.

The deterministic amplitude f_α and stochastic power q_α are not interchangeable and are not represented by one common parameter.

4 Modewise stability

Theorem 4.1 (Deterministic input-to-state bound). *Assume Assumption 2.1 and $\gamma_\alpha > 0$. Every solution of Equation (3.1) obeys*

$$|a_\alpha(t)| \leq e^{-\gamma_\alpha t} |a_\alpha(0)| + \frac{f_\alpha}{\gamma_\alpha} (1 - e^{-\gamma_\alpha t}),$$

and hence

$$\limsup_{t \rightarrow \infty} |a_\alpha(t)| \leq \frac{f_\alpha}{\gamma_\alpha}.$$

Proof. The upper Dini derivative satisfies $D^+|a_\alpha| \leq -\gamma_\alpha|a_\alpha| + f_\alpha$. Comparison with the scalar affine equation gives the result. $\square$

Theorem 4.2 (Stochastic second-moment bound). *Assume Assumption 2.1 and $\gamma_\alpha > 0$. Every solution of Equation (3.2) satisfies*

$$\frac{d}{dt}\mathbb{E}|a_\alpha|^2 \leq -2\gamma_\alpha\mathbb{E}|a_\alpha|^2 + q_\alpha,$$

and therefore

$$\mathbb{E}|a_\alpha(t)|^2 \leq e^{-2\gamma_\alpha t}\mathbb{E}|a_\alpha(0)|^2 + \frac{q_\alpha}{2\gamma_\alpha}(1 - e^{-2\gamma_\alpha t}).$$

Proof. Apply Itô's formula to $|a_\alpha|^2$, use the one-sided remainder bound, and take expectations. The martingale term has zero expectation and the quadratic variation contributes $q_\alpha dt$. $\square$

These two theorems are sufficient stability bounds for nonlinear dynamics. They do not assert that a nonlinear invariant law is Gaussian or that $\gamma_\alpha > 0$ is necessary for every particular forcing history.

5 Exact OU theorem and robust necessity

Theorem 5.1 (Exact scalar OU classification). *For*

$$da = -\gamma a dt + \sqrt{q} dW_t, \quad q > 0,$$

there is a unique invariant probability law with finite variance if and only if $\gamma > 0$. It is Gaussian with mean zero and variance

$$\text{Var}(a) = \frac{q}{2\gamma}.$$

For $\gamma = 0$ the variance grows linearly, and for $\gamma < 0$ it grows exponentially.

Theorem 5.2 (Worst-case deterministic sign criterion). *The scalar family $\dot{a} = -da + R(a) + b(t)$ with $\text{Re}(\bar{a}R(a)) \leq L|a|^2$ is uniformly input-to-state stable for every bounded input and every admissible remainder only if $\gamma = d - L > 0$. For a specified forcing/remainder pair, $\gamma \leq 0$ does not by itself prove divergence because cancellation or a vanishing input may occur.*

Thus “if and only if” belongs to exact OU dynamics or a declared robust worst-case class, not to arbitrary nonlinear stochastic equations.

6 Bundlewise noncoherent stability

Under bounded geometry and the joint spectral calculus, assume

$$c_1 \sum_{\alpha \in \mathcal{I}_Q} (1 + \Lambda_\alpha)|a_\alpha|^2 \leq \|Q\Psi\|_{H_F^1}^2 \leq c_2 \sum_{\alpha \in \mathcal{I}_Q} (1 + \Lambda_\alpha)|a_\alpha|^2. \quad (6.1)$$

6.1 Stochastic trace condition

For independent exact OU modes define

$$\Sigma_q := \sum_{\alpha \in \mathcal{I}_Q} (1 + \Lambda_\alpha) \frac{q_\alpha}{2\gamma_\alpha}.$$

If every $\gamma_\alpha > 0$ and $\Sigma_q < \infty$, then

$$\limsup_{t \rightarrow \infty} \mathbb{E} \|Q\Psi(t)\|_{H_F^1}^2 \leq c_2 \Sigma_q.$$

For the independent exact OU product this condition is also necessary for a finite- H_F^1 second moment. For nonlinear dynamics it remains a sufficient bound unless additional lower estimates establish necessity. With correlated noise the correct replacement is the corresponding weighted covariance trace.

6.2 Deterministic weighted-series condition

Define

$$\Sigma_f := \sum_{\alpha \in \mathcal{I}_Q} (1 + \Lambda_\alpha) \frac{f_\alpha^2}{\gamma_\alpha^2}.$$

If every $\gamma_\alpha > 0$ and $\Sigma_f < \infty$, then

$$\limsup_{t \rightarrow \infty} \|Q\Psi(t)\|_{H_F^1}^2 \leq c_2 \Sigma_f.$$

This is an amplitude-series condition, not a stochastic trace.

6.3 Weyl-law check

If the joint counting function satisfies $N(\Lambda) \lesssim \Lambda^{d_{\text{eff}}/2}$ and $\gamma_\alpha \gtrsim 1 + \Lambda_\alpha$, then $q_\alpha \lesssim (1 + \Lambda_\alpha)^{-p}$ with $p > d_{\text{eff}}/2$ is sufficient for $\Sigma_q < \infty$. The exact exponent must be recomputed if the damping growth or joint multiplicities differ.

No statement in this section controls a disturbance acting directly in $P\Psi$. Such coherent forcing needs a separate base/coherent stability theorem.

7 Deterministic homogenization of coherent dynamics

Write $\Psi = X + Y$ with $X = P\Psi$ and $Y = Q\Psi$. For frozen x , let the fast flow for Y have invariant measure μ_x , and decompose

$$g(x, y) = \bar{g}(x) + G(x, y), \quad \bar{g}(x) = \int g(x, y) d\mu_x(y).$$

The correctly scaled slow-fast system is

$$\dot{X}_\varepsilon = f(X_\varepsilon) + \bar{g}(X_\varepsilon) + \varepsilon^{-1/2} G(X_\varepsilon, Y_\varepsilon), \tag{7.1}$$

$$\dot{Y}_\varepsilon = \varepsilon^{-1} L_{X_\varepsilon}(Y_\varepsilon). \tag{7.2}$$

For fixed x define the stationary covariance

$$R_x(s) = \int G(x, Y_s) \otimes G(x, Y_0) d\mu_x(Y_0).$$

Assumption 7.1 (Enhanced invariance principle). On the selected compact slow domain:

- (H1) the frozen fast flow has a unique invariant measure and sufficient mixing for an integrable covariance;
- (H2) the centered additive functionals satisfy a functional CLT uniformly in x , and their laws are tight in the required path topology;
- (H3) the lifted first and second iterated integrals converge to a geometric Brownian rough path, with zero area anomaly for the canonical Stratonovich statement below;
- (H4) the Poisson equation/corrector and its x -derivatives have the bounds needed to control slow dependence; and
- (H5) initial layers and exits from the compact slow domain are controlled on $[0, T]$.

For infinite-dimensional coherent spaces, the corresponding Hilbert-space tightness, trace, and rough-path assumptions must be supplied separately.

Theorem 7.2 (Conditional homogenized limit). *Under Assumption 7.1, X_ε converges in law on $C([0, T])$ to*

$$dX_t = (f(X_t) + \bar{g}(X_t)) dt + \sigma(X_t) \circ dW_t,$$

where

$$D(x) := \sigma(x)\sigma(x)^* = \int_0^\infty (R_x(s) + R_x(s)^*) ds. \quad (7.3)$$

If the enhanced limit has a nonzero area anomaly, an additional deterministic bracket drift must be included; if only an ordinary functional CLT is known, the Stratonovich conclusion is not established.

8 Fixed points versus invariant measures

For autonomous or τ -periodic deterministic forcing, the corrected FP-I/II Schauder or Darbo theorem applies after a deterministic invariant set is proved; the result is respectively a step-fixed or periodic point, with equilibrium promotion only under the FP-II Lyapunov condition.

For stochastic forcing, the solution defines a Markov semigroup. An invariant probability measure requires, for example, the Feller property, a Lyapunov moment bound, and tight Krylov–Bogoliubov averages. Uniqueness requires an additional irreducibility/coupling or contractivity argument. A stochastic invariant measure is not called a deterministic fixed point.

9 Examples and execution checklist

For a circle operator with eigenvalues m^2 and linear damping $\gamma_m = m^2 - L$, exact OU noise gives $\text{Var}(a_m) = q_m/[2(m^2 - L)]$, whereas bounded deterministic forcing gives $\limsup |a_m| \leq f_m/(m^2 - L)$. These are different floors.

For a compact Heisenberg nilmanifold in a noncollapsing left-invariant metric class, the first positive eigenvalue has a uniform lower bound. This supplies one ingredient in d_α but does not replace the joint-mode or summability checks.

For every concrete geometry:

1. construct the joint spectral index and multiplicities;
2. compute d_α and prove the one-sided constants L_α ;

3. list deterministic amplitudes f_α and stochastic covariance data q_α or Q_ξ separately;
4. check $\gamma_\alpha > 0$ and the appropriate Σ_f or covariance trace;
5. control coherent forcing independently; and
6. for homogenization, verify the enhanced invariance principle rather than only quoting mixing.

10 Conclusion

The damping-margin principle survives, but with precise scope. Positive γ_α gives deterministic input-to-state and stochastic second-moment bounds. Exact Gaussian invariant laws and sign necessity belong to exact OU or robust worst-case formulations. Bundlewise results concern $Q\Psi$ and require either a stochastic trace or deterministic weighted series. The homogenized Stratonovich equation is conditional on enhanced functional-CLT/rough-path data and uses the corrected Green–Kubo normalization. These distinctions prevent stochastic invariant measures from being conflated with deterministic fixed points.

A Exact OU variance computation

For $da = -\gamma a dt + \sqrt{q} dW_t$,

$$a(t) = e^{-\gamma t} a(0) + \sqrt{q} \int_0^t e^{-\gamma(t-s)} dW_s.$$

Itô isometry gives

$$\mathbb{E} \left| \sqrt{q} \int_0^t e^{-\gamma(t-s)} dW_s \right|^2 = \frac{q}{2\gamma} (1 - e^{-2\gamma t})$$

when $\gamma > 0$.

B Uniform spectral gap for noncollapsing nil fibers

Proposition B.1. *Let $F = \Gamma \backslash \text{Nil}_3$ be fixed and let $\mathcal{G}_\epsilon$ be a compact class of left-invariant metrics satisfying $\epsilon I \leq G \leq \epsilon^{-1} I$. Then*

$$\inf_{g \in \mathcal{G}_\epsilon} \lambda_1(F, g) > 0.$$

The same bound is uniform for a smoothly base-dependent family remaining in $\mathcal{G}_\epsilon$.

Proof. The first positive eigenvalue varies continuously on this compact metric family and is positive for every compact connected fiber, so it attains a positive minimum; compare the spectral-geometric background in [3, 2, 7, 5]. $\square$

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